

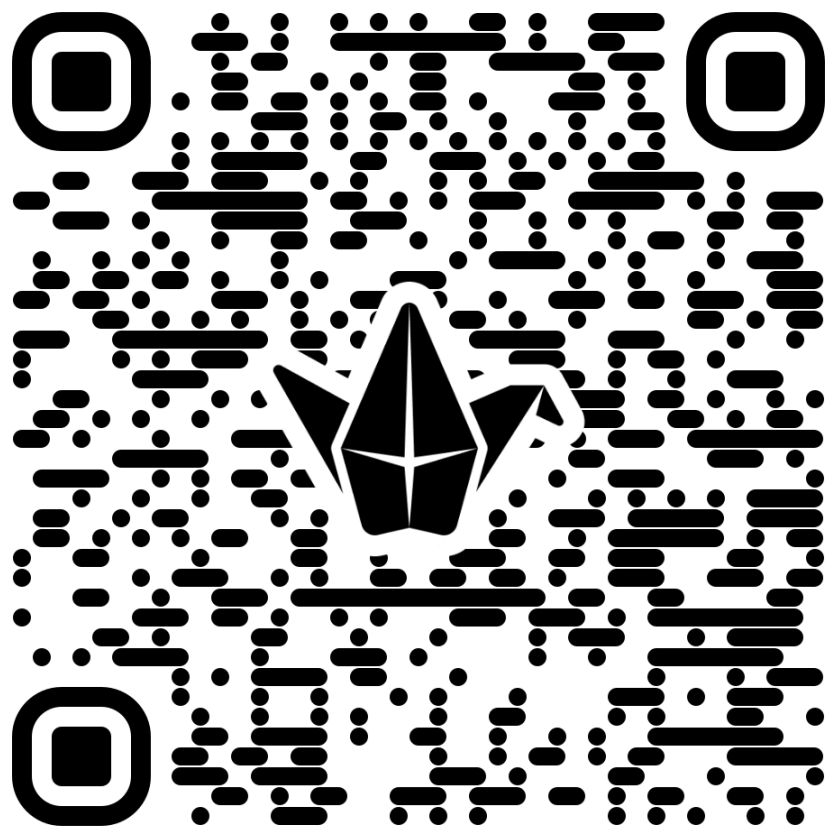
BIP

Hack Your Device, Fundamentals of Cyber Security

17 April – 15 May 2026



Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Education and Culture Executive Agency (EACEA). Neither the European Union nor EACEA can be held responsible for them. Grant Agreement Number: 101124674.



Camera

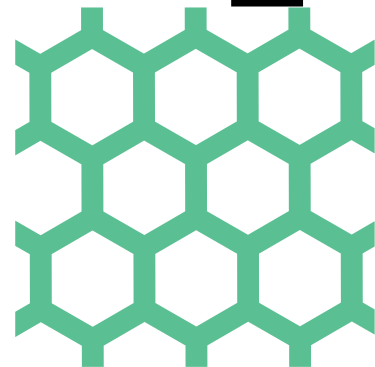
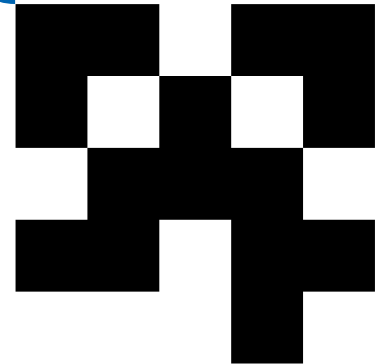
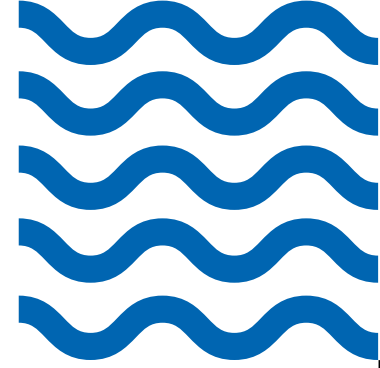
```
camera_is_on = False
```

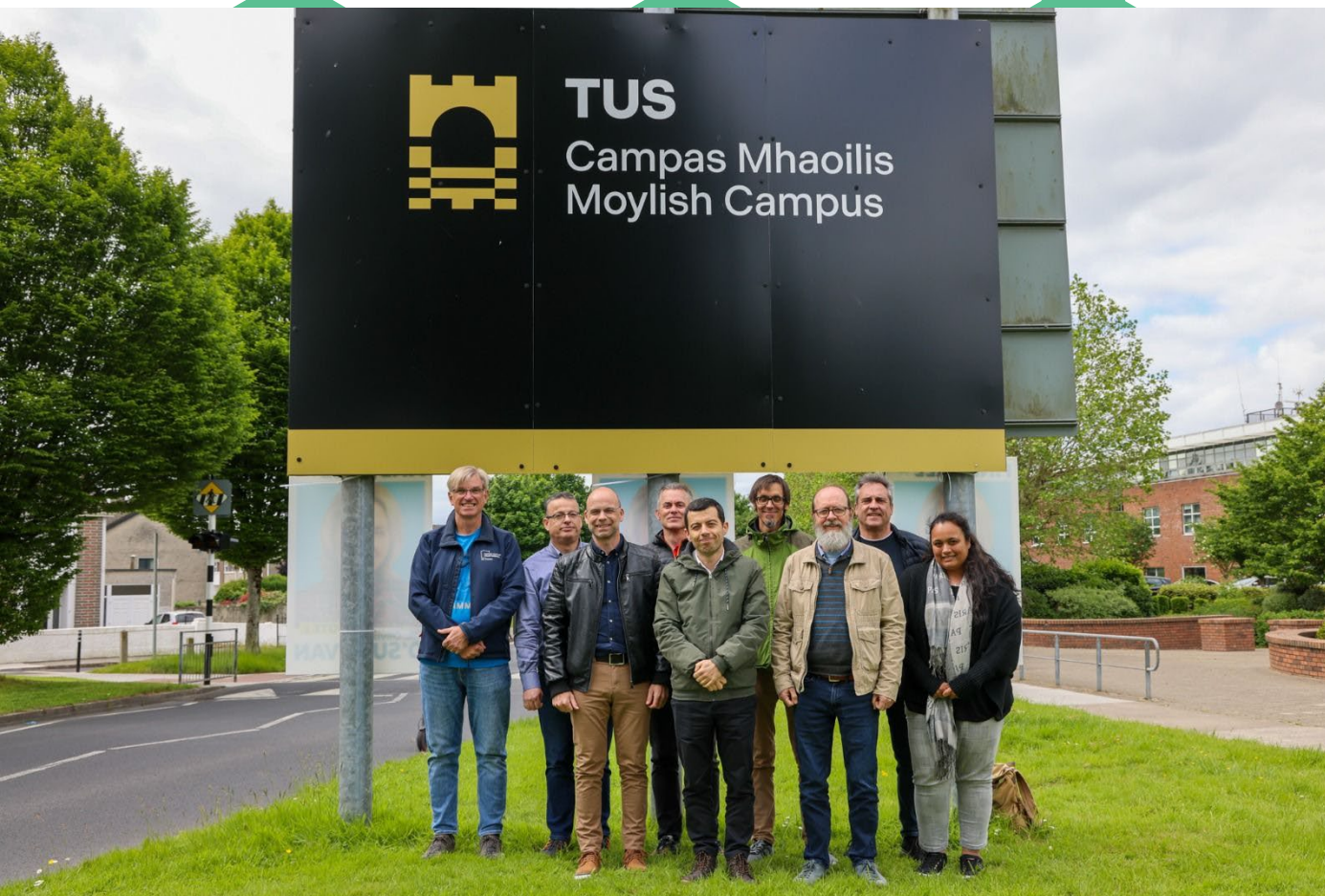
```
if not camera_is_on:  
    print("Camera is off... Let's turn it on! 🔄 ")  
    camera_is_on = True  
    print("I am present!")  
else:  
    print("Camera is already on! I am present")
```

Whatsapp group



edu.nl/cja43





Nice to meet you!

- Stephen McCombie (NHL Stenden)
- Ken Oakley (TUS)
- Leonel Santos (Polytechnic of Leiria) – Course Leader
- Nuno Lopes (IPCA)
- Armin Simma (FHV)
- Mike Winterburn (TUS)
- Jaqueline Berghout (NHL Stenden)
- Ismo Turve (Häme University of Applied Sciences)



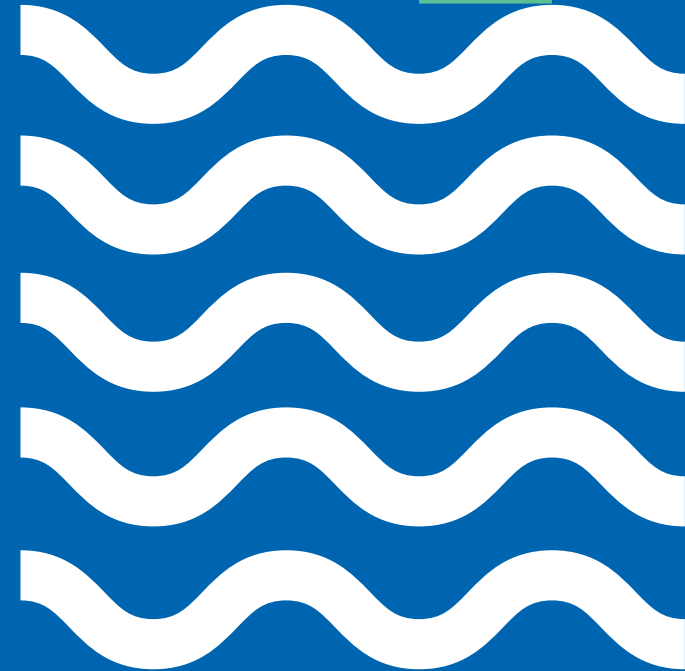
About last time:

Creating a European education
area

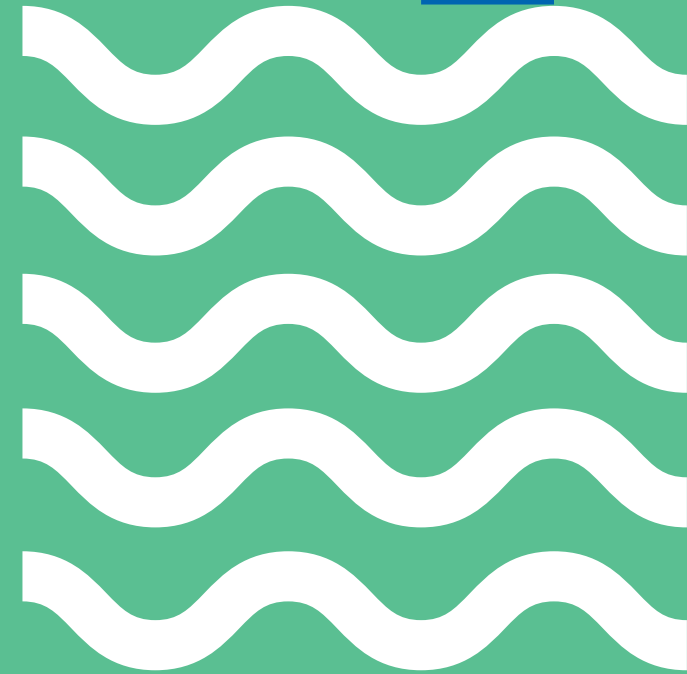
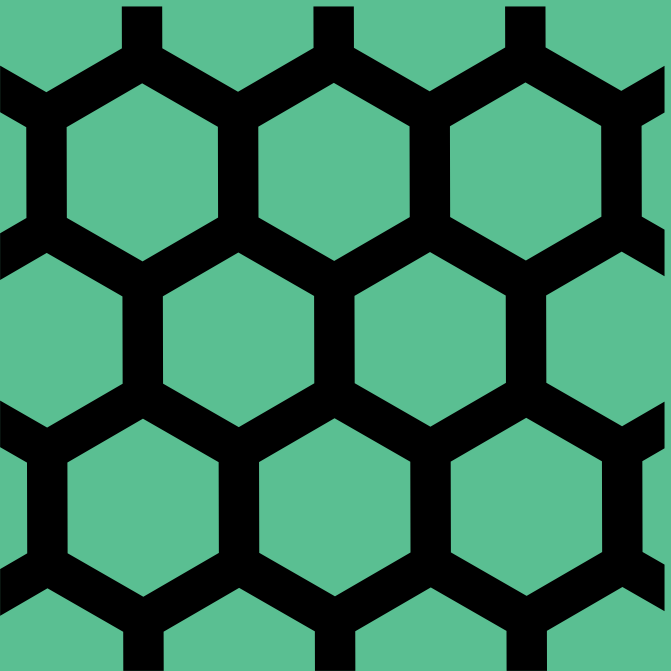
Strengthening regional
development

Developing future and advanced
skills

Unofficial result

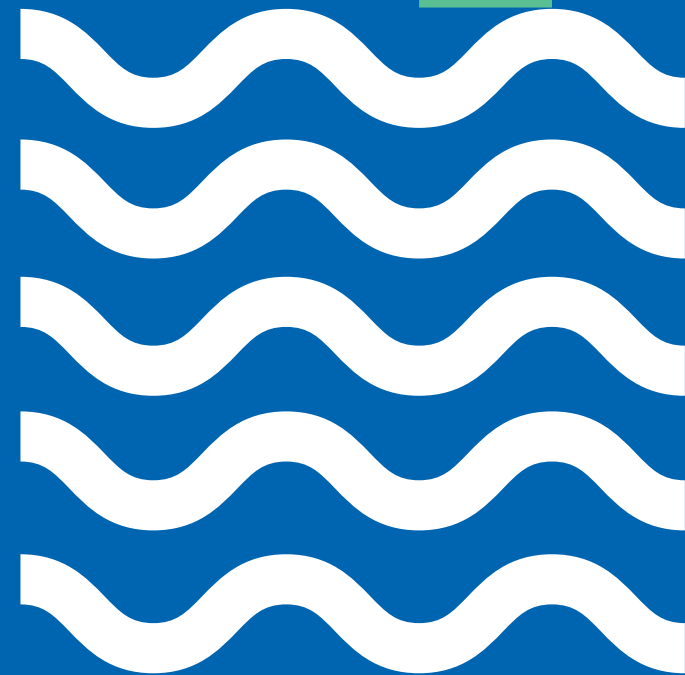


Friendship for life!

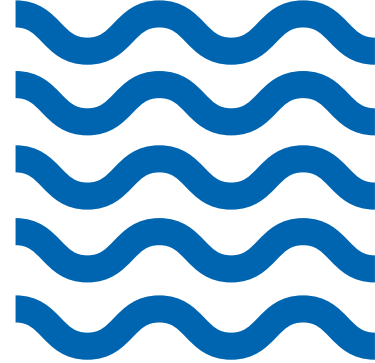




About this time



Politécnico de Leiria



Portuguese



Hello	Olá
Good morning	Bom dia
Goodbye	Adeus (formal and definitive) / Até logo (see you later)
Thank you	Obrigado/Obrigada
You're welcome	De nada
Excuse me	Desculpe/Desculpa
Please	Por favor

Portuguese

A beer, please!

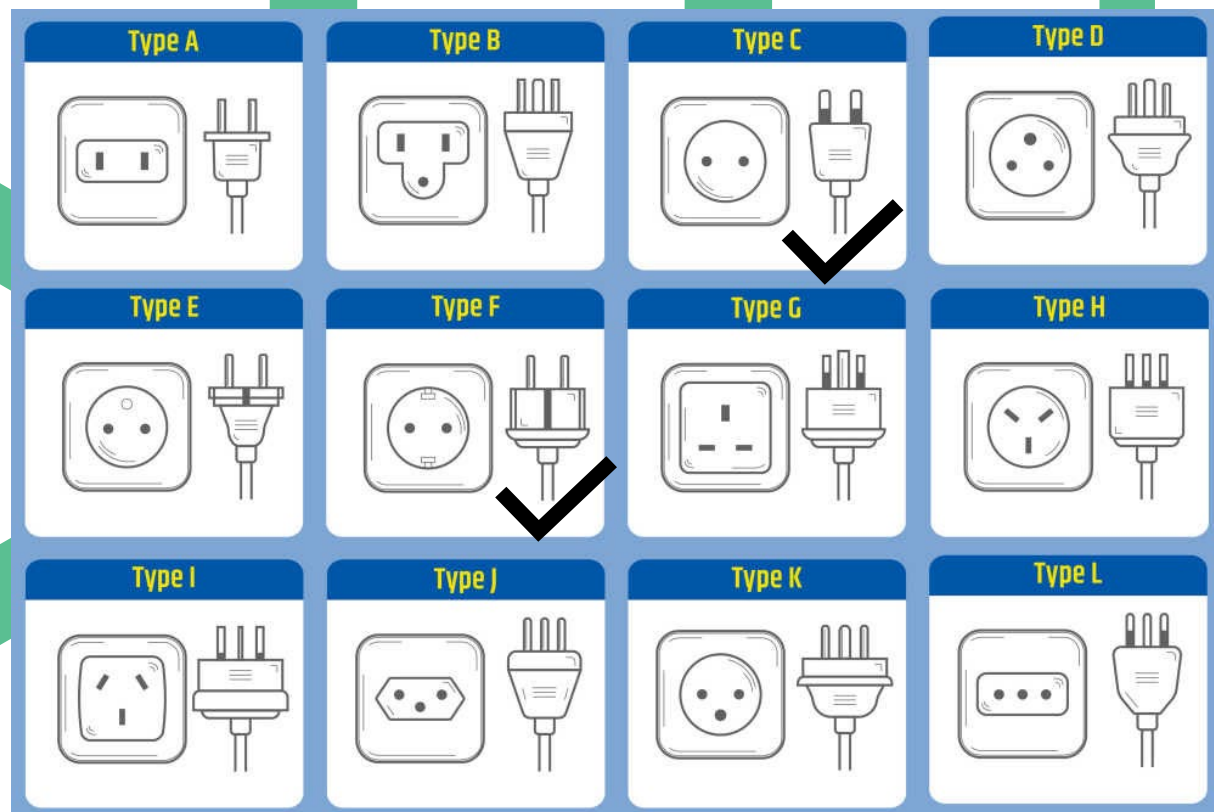
Uma cerveja, por favor 🍺

How much does it cost?

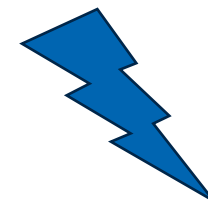
Quanto custa?

Where is the train station?

Onde fica a estação de comboios?



PLUGS



Type F: This is the most commonly used plug. It has two round pins and metal clamps on the side for grounding.

Type C: This is a plug with only two round pins without grounding, often used for low-power devices such as phone chargers.

Voltage: 230V

Frequency: 50Hz

CASH



In case of an emergency

<https://www.sns24.gov.pt/>

808 242424

On Friday 17 April 2026 there is a one-hour online session starting at 15:00 CEST where students get information about organisational matters (e.g. rooms at IPLeiria) and information about software that should be downloaded in advance (since it is about 20 GB in size).

BIP "Hack Your Device, Fundamentals of Cyber Security" (3 ECTS)

Schedule Week One (F2F) – 20 - 24 April 2026

Building B on Campus 2 - School of Technology and Management (ESTG)
Polytechnic University of Leiria, Portugal

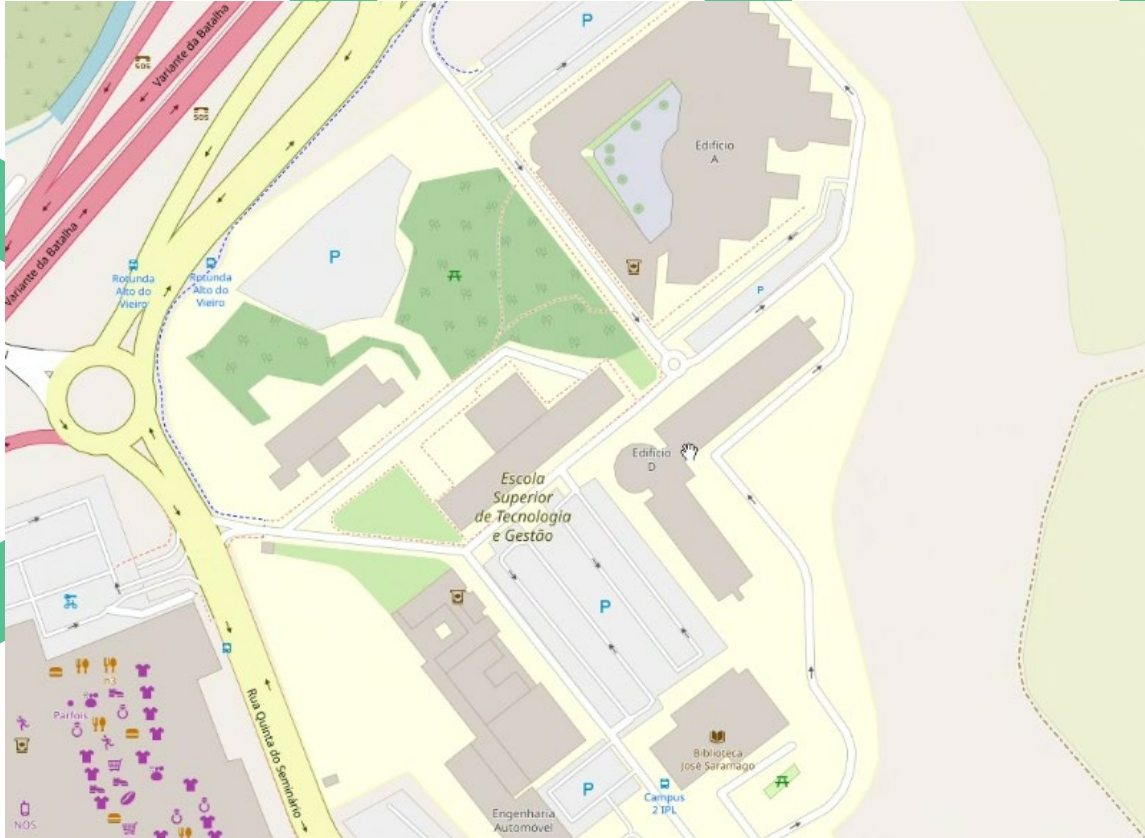
	9:00	9:30	10:00	10:30	11:00	11:30	12:00	12:30	13:00	13:30	14:00	14:30	15:00	15:30	16:00	16:30	17:00	17:30
MONDAY 20/4		9:30-10:30 Introduction, project outline, learning objectives		10:30-12:30 Basics of Cybersecurity & Networks				12:30-13:30 LUNCH BREAK		13:30-15:00 Groups/Team Formation/DBE/Intercultural Sensitivity			15:00-16:30 Setting up your hacking environment					
TUESDAY 21/4	9:00-11:00 Basic hacking (Lecture/Workshop)			11:00-12:30 IoT Getting started (Lecture/Workshop)				12:30-13:30 LUNCH BREAK		13:30-15:30 Pentesting IoT devices (Lecture/Workshop)				15:30-16:30 Public Speaking (Workshop/Lecture)				
WEDNESDAY 22/4			9:45-12:00 Cybersecurity and Industrial IoT visit (Lecture/Workshop)					12:00-14:00 LUNCH BREAK			14:00-18:00 Bus to Nazaré + Team Work @ Nazaré							
THURSDAY 23/4	9:00-11:00 First experiments with the device				11:00-12:30 Writing a Penetration Testing Report			12:30-13:30 LUNCH BREAK		13:30-16:00 Penetration Tests with the device				16:00- 16:30 Daily Reflection				
FRIDAY 24/4	9:00-12:30 Coaching Team Work and Coaching							12:30-13:30 LUNCH BREAK		13:30-15:30 Pentesting concept/Plan of Action Presentation (Public Presentation)				15:30-16:30 FAREWELL PARTY				

There are 15-minute breaks between sessions (not shown in this schedule).

Monday 20 April 2026

- 9 AM
- What if you are late?

Training Room 1 ("Sala de Formação 1")
located on the floor -1
Building B on Campus 2.



Requirements



20.0 GB Hard Disk Space – SSD preferred
Minimum of 2048 MB of RAM

Install a Desktop Hypervisor



For Windows Users: Use VirtualBox or VMware Workstation

- VMware Workstation Player (free for personal use):
<https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>
- VirtualBox (free, open-source):
<https://www.virtualbox.org/wiki/Downloads>

For Intel-based Macs: Use VirtualBox or VMware Fusion

- VMware Fusion (free for personal use on M1/M2):
<https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>
- VirtualBox (Not recommended for M1/M2 Macs, works well on Intel Macs):
<https://www.virtualbox.org/wiki/Downloads>

For Apple Silicon (M1/M2): Use QEMU or UTM

- UTM for Mac:
<https://mac.getutm.app/>
- QEMU:
<https://www.qemu.org/download/>

Download and Install Kali Linux as a Virtual Machine

<https://www.kali.org/get-kali/#kali-virtual-machines>



Pre-built Virtual Machines

Kali Linux [VMware](#) & [VirtualBox](#) images are available for users who prefer, or whose specific needs require a virtual machine installation.

These images have the [default credentials](#) "kali/kali".

[Virtual Machines Documentation >](#)

Recommended



VMware



3.2G

torrent

docs

sum

Recommended



VirtualBox



3.2G

torrent

docs

sum

Recommended



Hyper-V



3.2G

torrent

docs

sum

Recommended



QEMU



3.2G

torrent

docs

sum

While you are downloading

- WiFi Pineapple
- Rubber Ducky
- Kali Linux



Ethics

- *These materials can be bought anywhere. They are not illegal. However, deploying them may be illegal. And apart from whether or not something is allowed by the law, consideration should be given to whether something is ethical*
- *Also remember that these tools should not be used for unethical or illegal behaviour. You only deploy these resources on your own laptop and smartphone.*

In short

We obey t

Liudger Drachten doet aangifte na datalek: leerlingen hackten docentenlaptop

24 januari 2022, 17:36 • 1 minuut leestijd

Twee leerlingen van CSG Liudger Raai in Drachten hebben de persoonlijke gegevens van 1.150 leerlingen, hun ouders en medewerkers verspreid. De scholengemeenschap heeft aangifte gedaan van het lek.



© Omrop Fryslân, Darshan Boerema

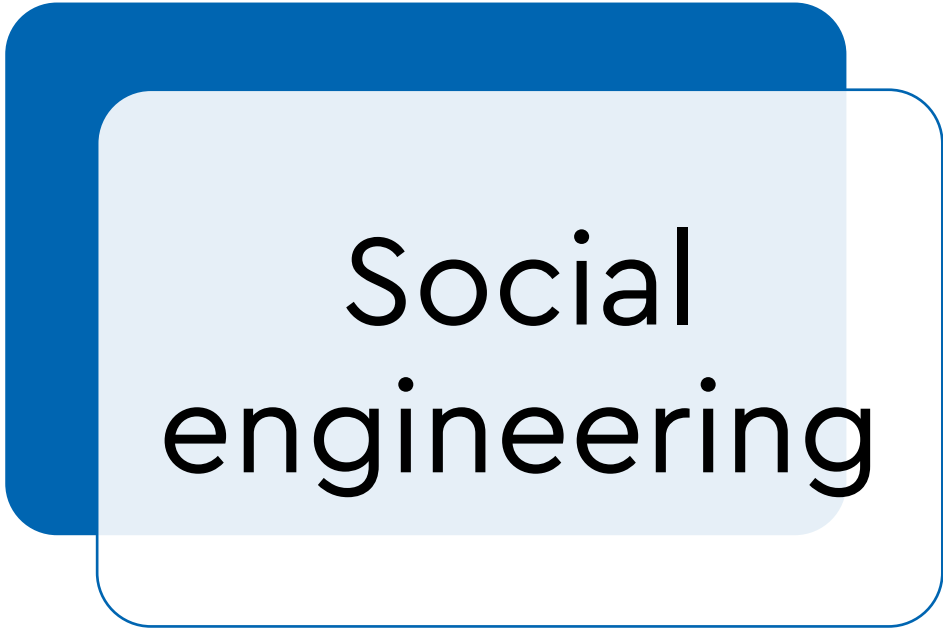
De betrokkenen zijn maandagmiddag op de hoogte gebracht van het lek. Ook de Autoriteit Persoonsgegevens is op de hoogte gebracht. Dat laat een woordvoerder van de school weten.

Software

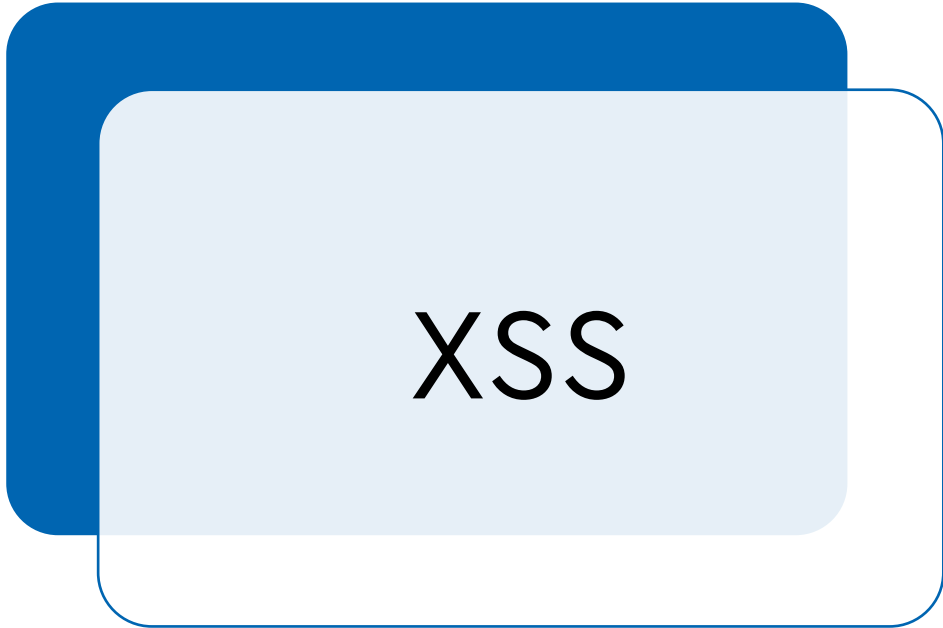
Eind november mocht een leerling een laptop van een docent gebruiken, omdat hij zelf niet een bij zich had. De leerling heeft toen een programma geïnstalleerd op de computer, waardoor hij samen met zijn schoolgenoot toegang had tot de gegevens van de laptop.

Beide leerlingen zijn op dit moment geschorst.

How hard is it? For example:

A graphic consisting of a dark blue rounded rectangle with a light blue rounded rectangle inside it, offset to the right and bottom.

Social
engineering

A graphic consisting of a dark blue rounded rectangle with a light blue rounded rectangle inside it, offset to the right and bottom.

XSS



CIA in IT, OT & IoT

How it differs per environment

IT



IT

Confidentiality First

-  Protect data
-  No data changes
-  Systems working

✓ **Data First**

OT



OT

Availability First

-  Systems running
-  Process correct
-  Privacy less

✓ **Continuity First**

IoT



IoT

All Important

-  Weak security
-  Leak & tamper
-  Disable device

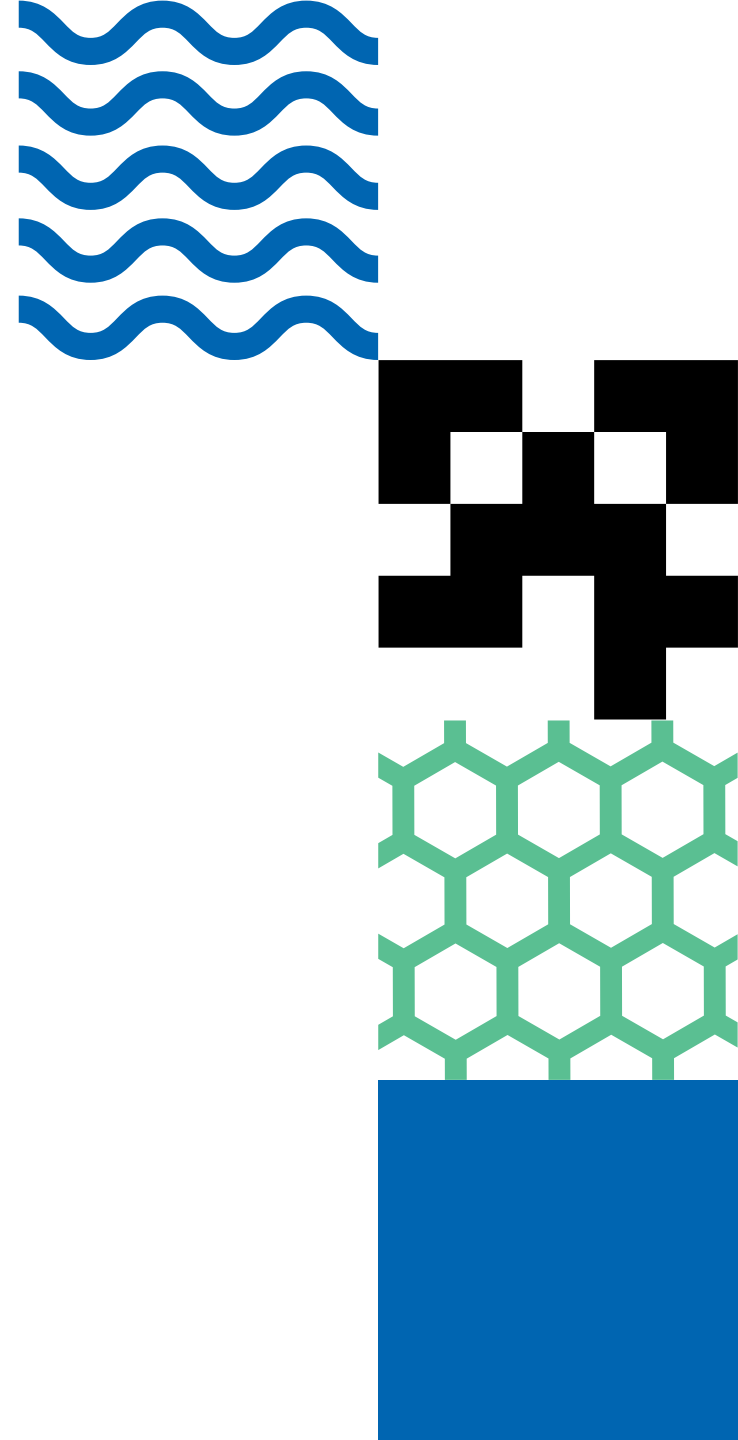
✓ **All Risk**

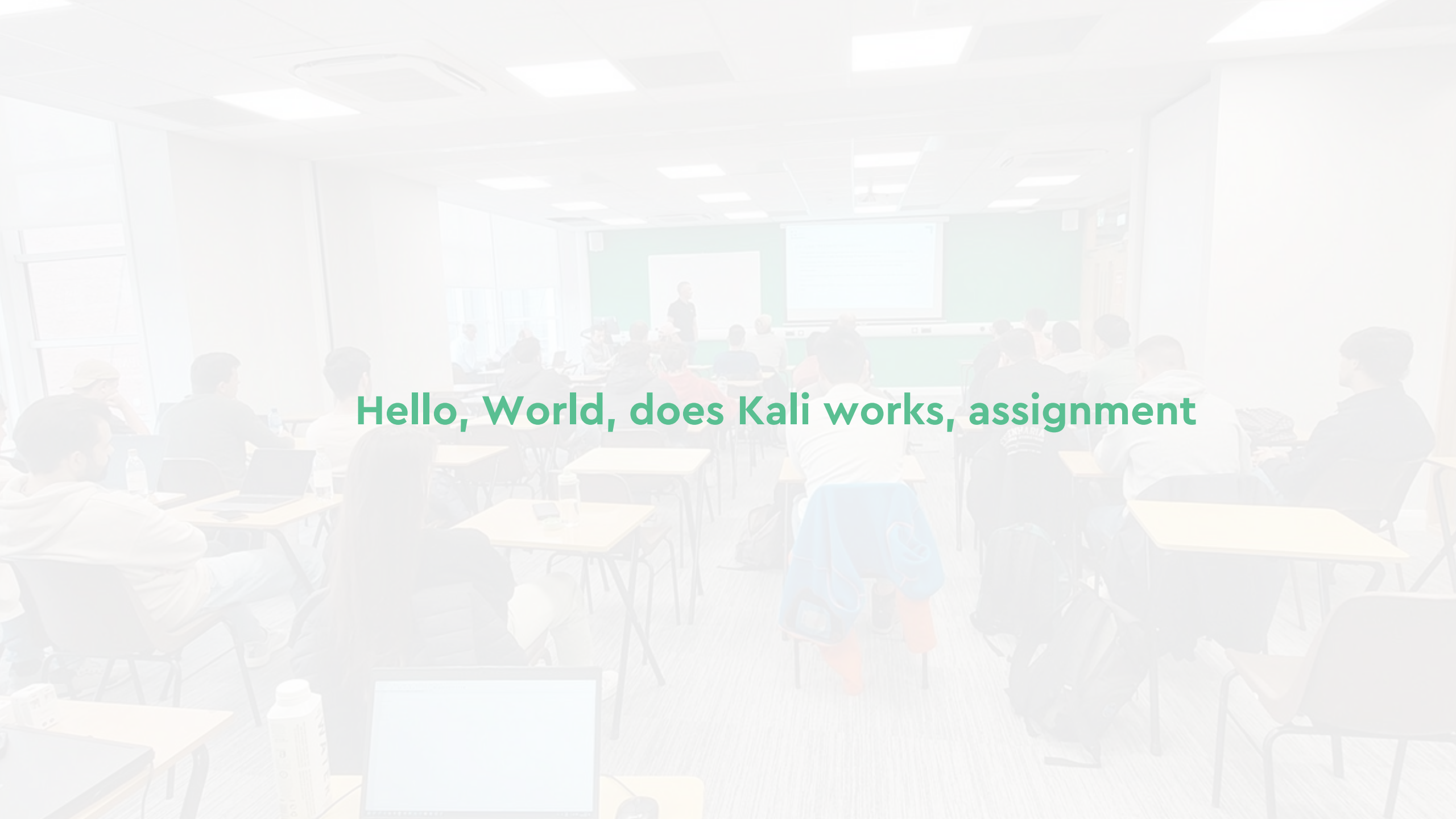
Mr Robot - Social Engineering - YouTube

kali - kali

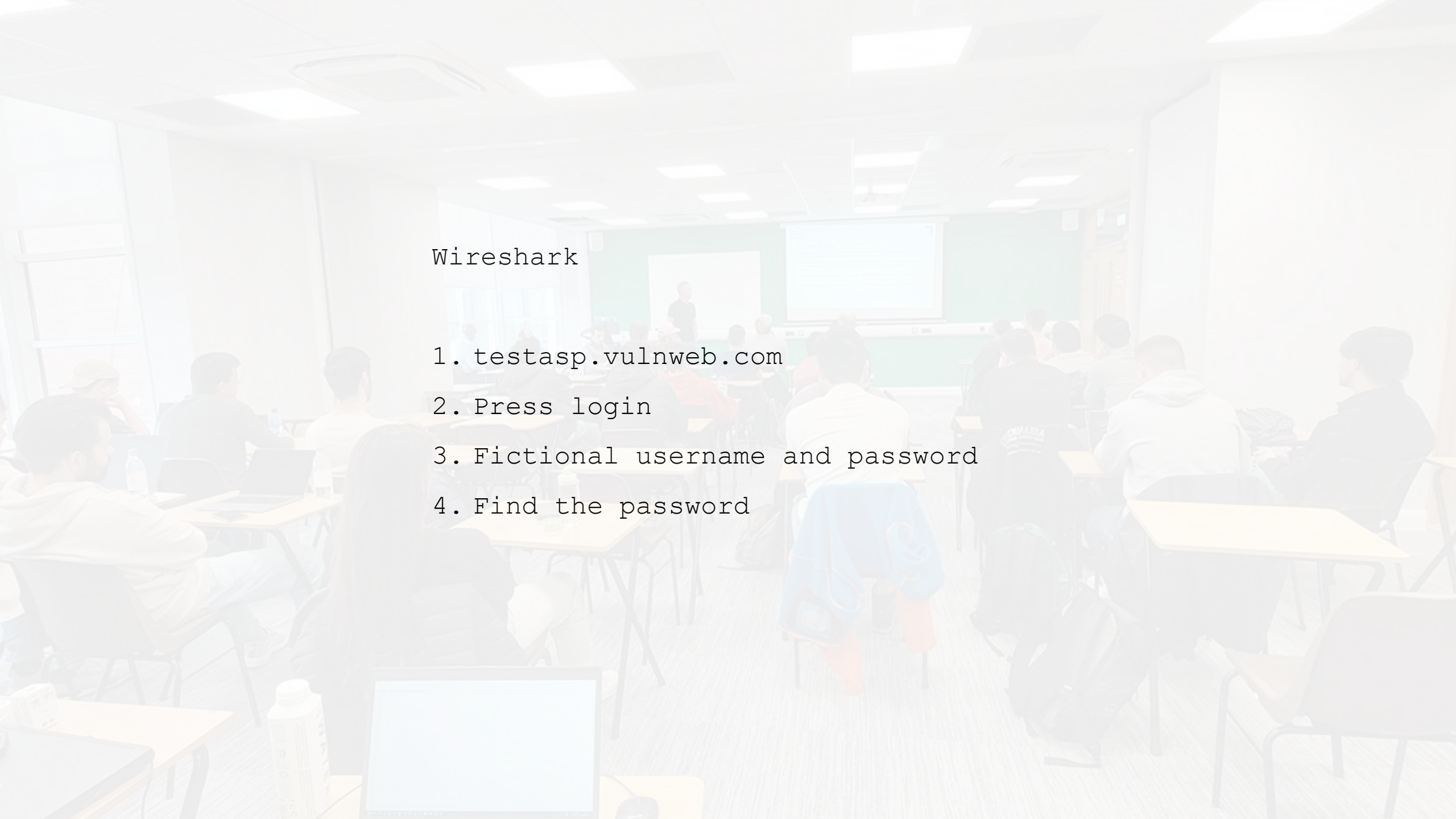
Errors that may occur:

- Windows Hyper-V Conflicts with VirtualBox
bcdedit /set hypervisorlaunchtype off
- Vboxsup not running
sc query vboxsup
sc start vboxsup





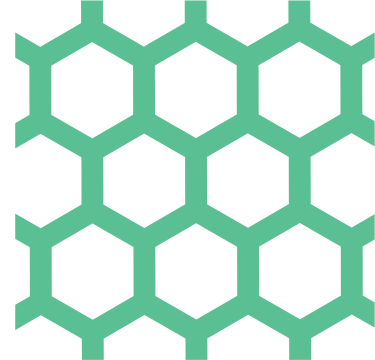
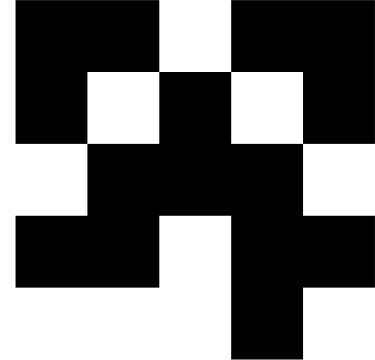
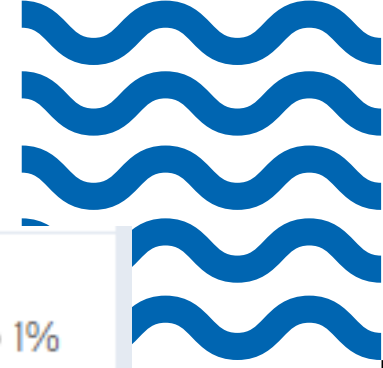
Hello, World, does Kali works, assignment



Wireshark

1. `testasp.vulnweb.com`
2. Press login
3. Fictional username and password
4. Find the password

Weather forecast



MON
4/20



24° 17°

Clouds giving way to some sun



Cloudy

☁ 1%

TUE
4/21



25° 13°

Some sun; breezy in the p.m.



Mostly cloudy

☁ 5%

WED
4/22



21° 10°

Some sun



Partly cloudy

☁ 1%

THU
4/23



22° 11°

Sun through high clouds



Becoming cloudy

☁ 25%

FRI
4/24



21° 14°

Decreasing clouds



A couple of showers late

☁ 4%

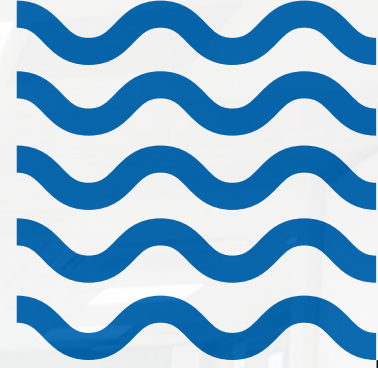


Pictures??

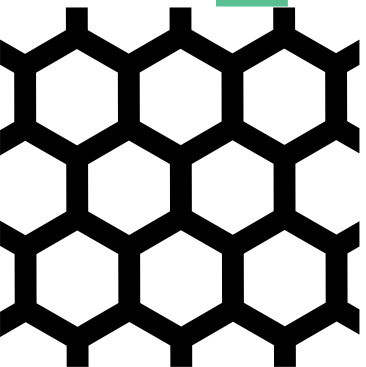
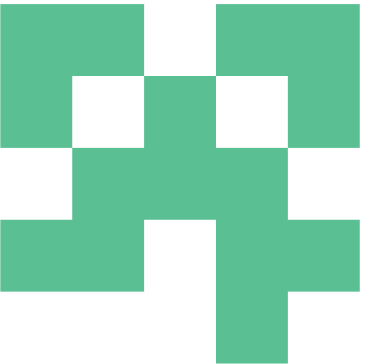
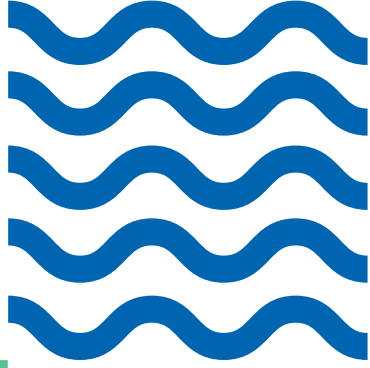
LinkedIn

Whatsapp Group

Instagram



Quiz:



What is the one thing we need to remember?

What is the abbreviation of BIP?

What was the name of the hacker in the true story of Sinterklaas?

How do you order a beer in Portuguese?

What is the one thing we need to remember?

Há alguma
pergunta?

